GS BASIC COURSES FOR POSTGRADUATES

DATA SCIENCE IN SOCIETY 5.0

1ST LESSON

Professor, Vice Director,
Center for Management of Teaching and Learning
Toru Hayashi

WHAT ARE GS BASIC COURSES FOR POSTGRADUATES?

STRUCTURE AND REQUIREMENTS OF GS BASIC COURSES FOR POSTGRADUATES

科目区分 Subjects Category		授業科目の名称 Subjects	英文科目名 Subjects in English Title	単位数 Credits		備考 Note
				必修 Requirement	選択 Elective	
研究科共通科目 Graduate School Common Courses	大学院ＧＳ基盤科目 GS Basic Courses for Postgraduates	異分野研究探査Ⅰ	Laboratory RotationⅠ	0.5		
		異分野研究探査Ⅱ	Laboratory RotationⅡ	0.5		
		研究者倫理	Research Ethics	1		
		知識集約型社会とデータサイエンス	Data Science and Society 5.0		1	選択必修1単位以上 Required to take more than 1 credit.
		次世代の先端科学技術	Advanced Science and Technology in the Next Generation		1	
		スマート創成科学	Smart Science and Technology for Innovation		1	
		イノベーション方法論Ａ	Innovation Methodology A		1	
		イノベーション方法論Ｂ	Innovation Methodology B		1	
		数理・データサイエンス・ＡＩ基盤	Mathematical, Data Science, and AI Basic		1	
		人間と社会の課題	Human and Social Challenges		1	選択必修1単位以上 Required to take more than 1 credit.
		ビジネス・技術マネジメント戦略論	Strategy for Business and Technology Management		1	
		ヘルスケア・イノベーション	Innovation in Healthcare		1	
		破壊的イノベーションに向けた技術経営論	MoT as for Disruptive Innovation		1	

GROUND RULES FOR TAKING GS BASIC COURSES FOR POSTGRADUATES "DATA SCIENCE IN SOCIETY 5.0”

IN CHARGE OF THE COURSE "DATA SCIENCE IN SOCIETY 5.0" IN GS BASIC COURSES FOR POSTGRADUATES

Toru Hayashi,
Center for Management of Teaching and Learning

Hidetaka Nambo,
Faculty of Transdisciplinary Sciences for Innovation,
Institute of Transdisciplinary Sciences for Innovation

AIM OF "DATA SCIENCE IN SOCIETY 5.0” AS GS BASIC COURSES FOR POSTGRADUATES

Subject matter of the class, etc.

In order to practice diverse and challenging research activities in a knowledge-intensive society where new normal life is required, students will learn the value of STEAM education that is oriented toward fusion of humanities and sciences and fusion of disciplines, and the future of DX useful for data-driven research, so that they can maximize their own areas of expertise, achievements, etc.

This class provides an opportunity to look at one's own interests as a researcher and to take a step toward fostering the challenging nature of new research as a graduate student. Therefore, the objective is to understand the significance of comprehensive knowledge with a broad perspective and the ability to act, which is required in a knowledge-intensive society, while learning about the possibilities of research that transcends one's own field of expertise and advanced examples of data utilization and DX.

Student Learning Objectives

(1) To be able to explain the image of human resources required in a knowledge-intensive society.

(2) Explain the value of education that integrates arts and sciences and disciplines.

(3) Explain the value of data science.

(4) Explain how DX can be used and its potential.

LESSON PLAN OF "DATA SCIENCE IN SOCIETY 5.0"

講義回	テーマ	具体的な内容	担当教員
1	Recent science and technology policies and human resource development	On-demand materials will be utilized.	林　透[HAYASHI, Toru](教学マネジメントセンター)
2	Significance and value of STEAM education and integrated knowledge in a knowledge-intensive society	On-demand materials will be utilized.	林　透[HAYASHI, Toru](教学マネジメントセンター)
3	Overview of Artificial Intelligence and Data Science	On-demand materials will be utilized.	南保　英孝[NAMBO, Hidetaka](融合研究域 融合科学系)
4	Deep learning	On-demand materials will be utilized.	南保　英孝[NAMBO, Hidetaka](融合研究域 融合科学系)
5	Sparse modeling	On-demand materials will be utilized.	南保　英孝[NAMBO, Hidetaka](融合研究域 融合科学系)
6	Deep reinforcement learning and GANs	On-demand materials will be utilized.	南保　英孝[NAMBO, Hidetaka](融合研究域 融合科学系)
7	Career Survival and Career Design in Society 5.0	On-demand materials will be utilized.	林　透[HAYASHI, Toru](教学マネジメントセンター)
8	Qualitative Integration Workshop for Looking at your own Research	On-demand materials will be utilized.	林　透[HAYASHI, Toru](教学マネジメントセンター)

SUBMISSION AND GRADING OF ASSIGNMENTS

【1st-2nd Lesson (in charge of Prof. Toru Hayashi)】
For the first and second Lesson, please read the handout materials and listen to the on-demand videos for each Lesson and respond to the review report assignment for each Lesson by the designated due date. (20 points distribution)

【3rd-6th Lesson (in charge of Prof. Satoshi Yamane)】
From 3rd to 6th Lesson, please read the handout materials and listen to the on-demand videos d materials for the 4 Lessons in order and respond to one assignment report by the designated due date, summarized in the 4 Lessons.
(40 points distribution)

【7th-8th Lesson (in charge of Prof. Toru Hayashi)】
For 7th and 8th Lesson, read the handout materials and listen to the on-demand videos for each Lesson and respond to the review report assignment for each Lesson by the designated due date. (40 points distribution)

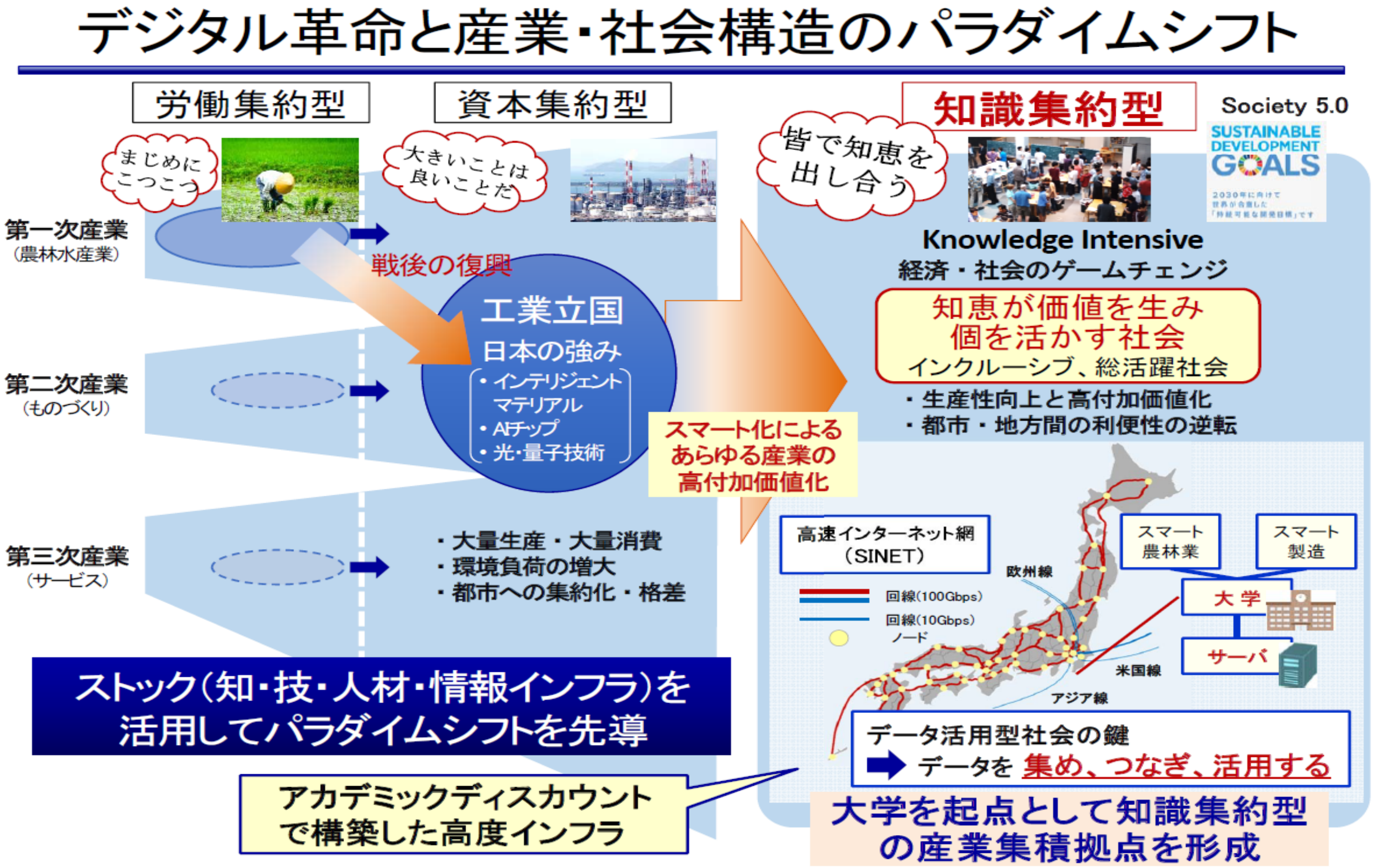
1ST LESSON THEME
"RECENT SCIENCE AND TECHNOLOGY POLICY AND HUMAN RESOURCE DEVELOPMENT"
PART 1: DIRECTION OF SCIENCE AND TECHNOLOGY POLICY

What is a Knowledge Intensive Society ➡ Society 5.0

The society of the future will not be an extension of the transition from labor-intensive to capital-intensive. The use of new information and communication technologies and data will lead to a discontinuous transformation of all industries, whether primary, secondary, or tertiary, into a new model. Such an industry model could be described as "knowledge-intensive.

Source: Makoto Gonokami (2019), "Future Map of Universities - Creating a 'Knowledge Intensive Society'", Chikuma Shinsho, pp. 47.

What is a Knowledge Intensive Society ➡ Society 5.0



[Source] Makoto Gonokami (2017), "Social Transformation toward Society 5.0 (Knowledge Intensive Society) and the Role of Universities."
Fiscal System Subcommittee of the Council on Fiscal System and Other Related Matters Materials (2017.10.4)

What is a Knowledge Intensive Society ➡ Society 5.0

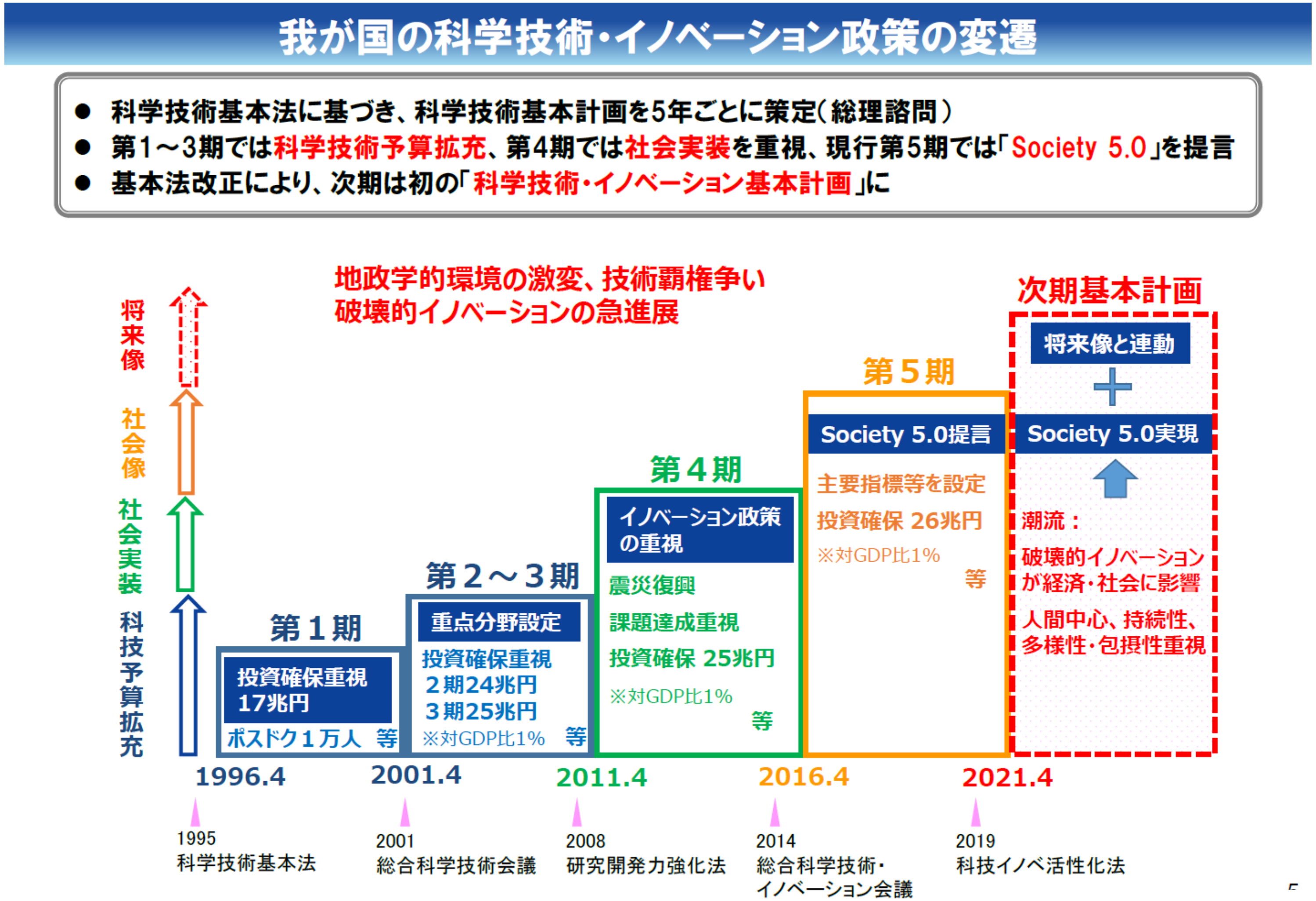
With the digital revolution and globalization, the world is undergoing a major transformation toward a knowledge-based society.

In light of this paradigm shift in the social system, in order for Japan, which has advocated "Society 5.0," to be a world leader in realizing this vision, we must quickly establish a new knowledge-intensive value creation system, solve social issues through science and technology innovation, and contribute to the sustainable development of the world, while at the same time, we must have a strong will to realize an inclusive society in which advanced science and technology are in harmony with society. At the same time, we must work with a strong will to realize an inclusive society where advanced science and technology and society are in harmony with each other.

To this end, we will strengthen the excellence and diversity of basic and academic research, which is the source of value creation, and universities (including Inter-University Research Institutes), which are the nucleus of a knowledge-intensive society. The same applies hereinafter). The expansion of the roles of universities (including Inter-University Research Institutes) and national research and development corporations, which are the core of a knowledge-intensive society, and the development of innovation leaders, are the most important issues to be addressed. To address these challenges, Japan must not hesitate to undertake intensive public-private investment in science and technology innovation and systemic reforms.

[Source] Council for Science, Technology and Science, Special Committee on General Policy (2020), "Science and Technology Innovation for Knowledge Intensive Value Creation.
Policy Development - Toward a World Leading Nation through the Realization of Society 5.0 (Final Summary)".

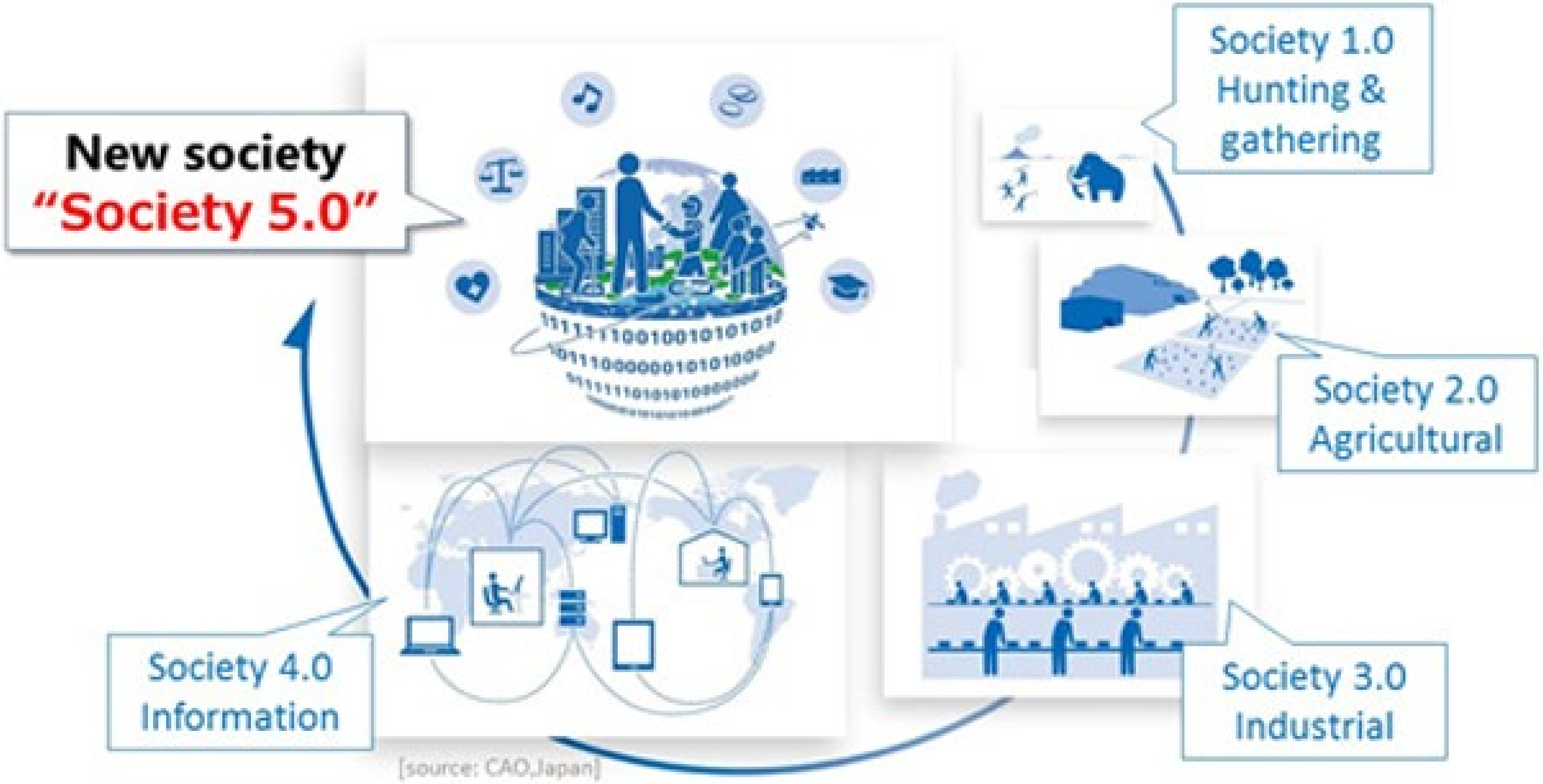
JAPAN'S SCIENCE, TECHNOLOGY AND INNOVATION POLICY "THE SIXTH SCIENCE, TECHNOLOGY AND INNOVATION BASIC PLAN" (FY2021-2025)



[Source] Director-General for Policy Coordination (Science, Technology and Innovation), Cabinet Office, Government of Japan (2020).

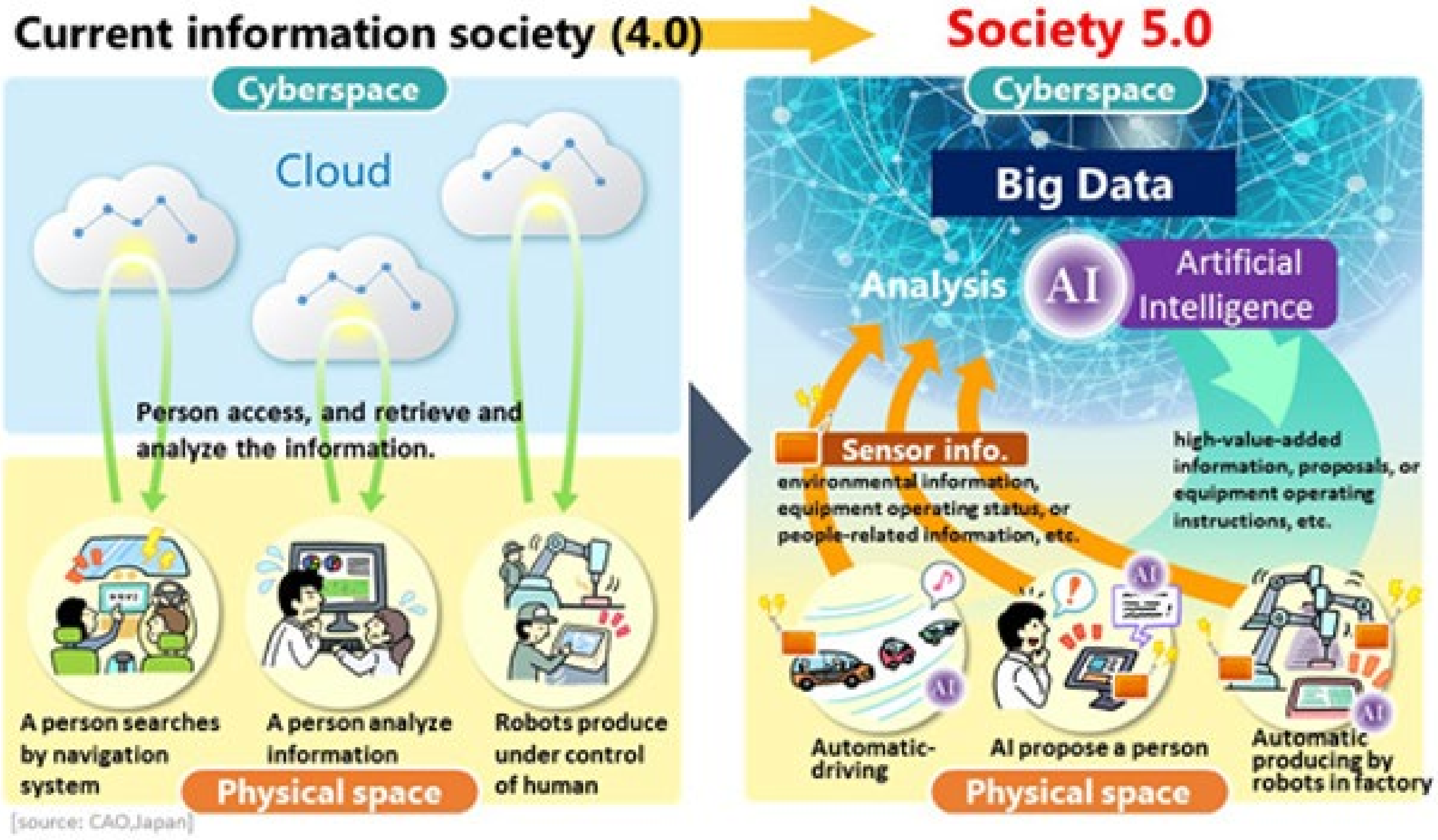
Basic Plan for Science, Technology and Innovation: Toward the Realization of Society 5.0

Society 5.0 Society



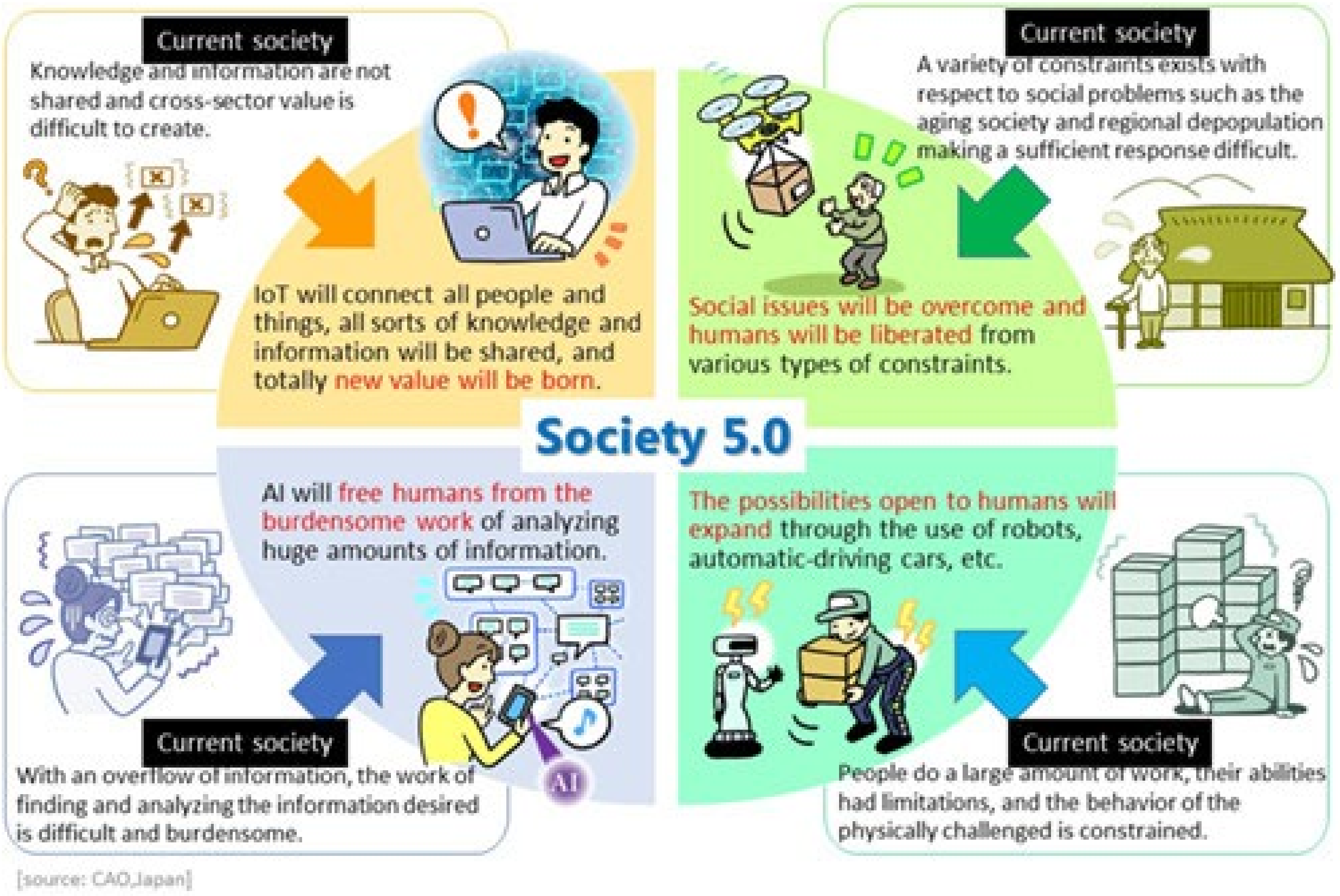
[Source] Director-General for Policy Coordination (Science, Technology and Innovation), Cabinet Office, Government of Japan (2020). Basic Plan for Science, Technology and Innovation: Toward the Realization of Society 5.0

Society 5.0 Society



[Source] Director-General for Policy Coordination (Science, Technology and Innovation), Cabinet Office, Government of Japan (2020). Basic Plan for Science, Technology and Innovation: Toward the Realization of Society 5.0

Society 5.0 Society



[Source] Director-General for Policy Coordination (Science, Technology and Innovation), Cabinet Office, Government of Japan (2020). Basic Plan for Science, Technology and Innovation: Toward the Realization of Society 5.0

Society 5.0 Society

[Source] Director-General for Policy Coordination (Science, Technology and Innovation), Cabinet Office, Government of Japan (2020). Basic Plan for Science, Technology and Innovation: Toward the Realization of Society 5.0

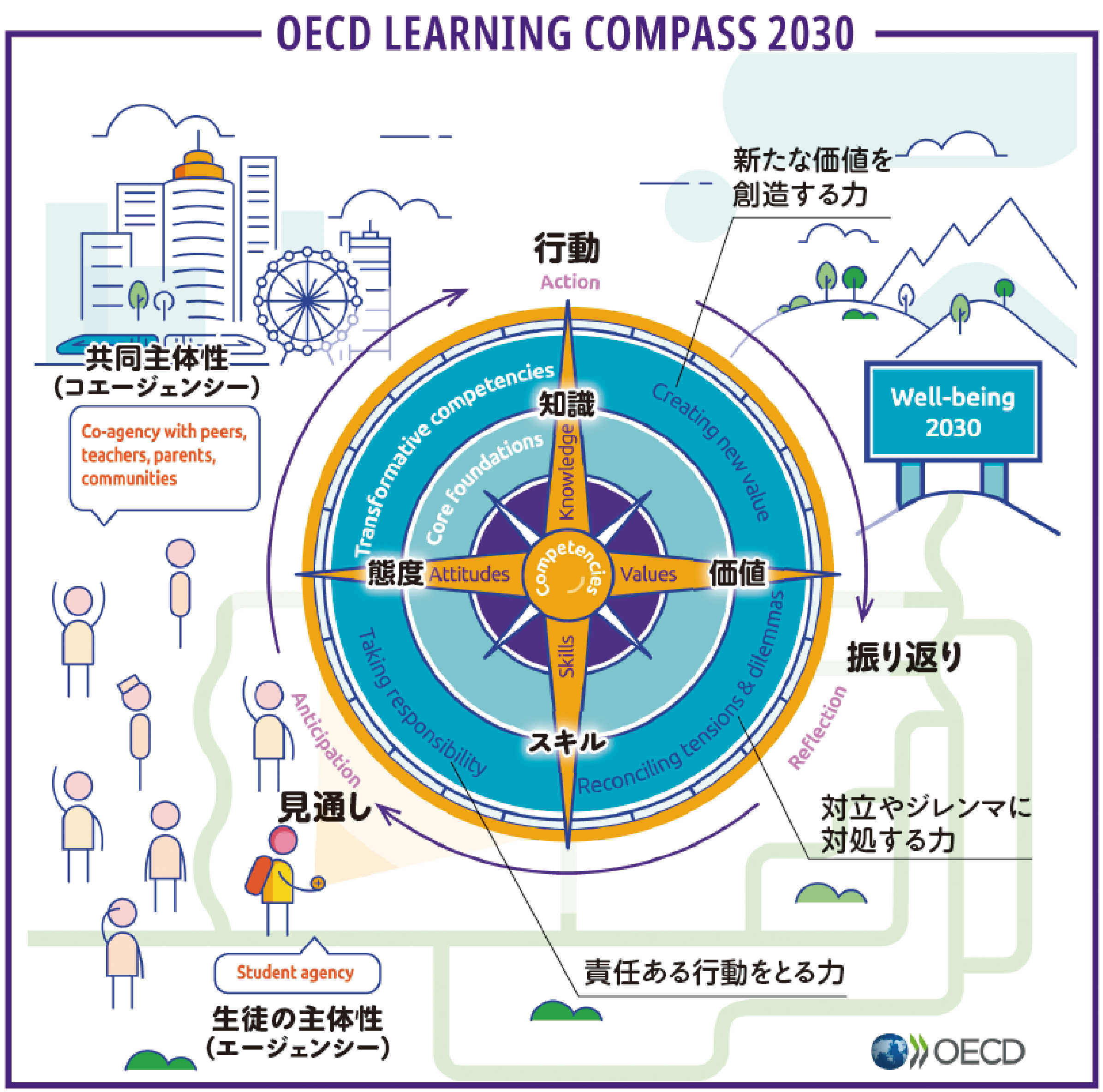
Japan Challenge for Society 5.0 : Accelerate Innovation with Japan

Video prepared by JETRO, Japan External Trade Organization (35 min. 24 sec.)

Please click on the URL below to watch the video.
https://youtu.be/oTBIiHPaqkM

1ST LESSON THEME
"RECENT SCIENCE AND TECHNOLOGY POLICY AND HUMAN RESOURCE DEVELOPMENT“
PART 2: HUMAN RESOURCE DEVELOPMENT

Direction of Human Resource Development for the 2030 Society



▲ OECD の Education2030 プロジェクトで創り上げた
ラーニング・コンパス（学びの羅針盤）

Source.
General Incorporated Association
Education Alliance Network
Issued February 19, 2021
Let us have the power to be a compass for the future!
(Koleima NO.13)

Skills needed in 2030 society

	USA
1	Learning Strategies
2	Psychology
3	Instructing
4	Social Perceptiveness
5	Sociology and Anthropology
6	Education and Training
7	Coordination
8	Originality
9	Fluency of Ideas
10	Active Learning

H Bakhshi, JM Downing, MA Osborne, P Schneider (2017) "The future of skills: employment in 2030"

https://futureskills.pearson.com/research/assets/pdfs/media-pack.pdf

FOSTERING LEADERS OF INNOVATION IN A KNOWLEDGE INTENSIVE SOCIETY

(1) Basic Direction

As society shifts to a knowledge-intensive society, the field of new value creation is shifting from "goods" to "things," and even start-ups and individuals without large-scale capital can create new value and innovations.

In such a society, it is important to have a system where individuality is converted into strengths and diverse values are tolerated, rather than where "the stakes are high.

[Source] Council for Science, Technology and Science, Special Committee on General Policy (2020), "Science and Technology Innovation for Knowledge Intensive Value Creation.
Policy Development - Toward a World Leading Nation through the Realization of Society 5.0 (Final Summary)".

From the Design of Objects to the Design of Things

コトのデザインとは？

「コトのデザイン」とは「モノのデザイン」と対を成す考え方です。一般的にデザインというと、物のスタイルや色合いを思い浮かべる方が多いのではないでしょうか。ところで世界的には、デザインの本質的な考え方やプロセスへの認識が進められているようです。Safeology研究所・研究員の谷内眞之助によるとデザインを「モノのデザイン」と「コトのデザイン」として捉え、違いを以下のように提案しています。

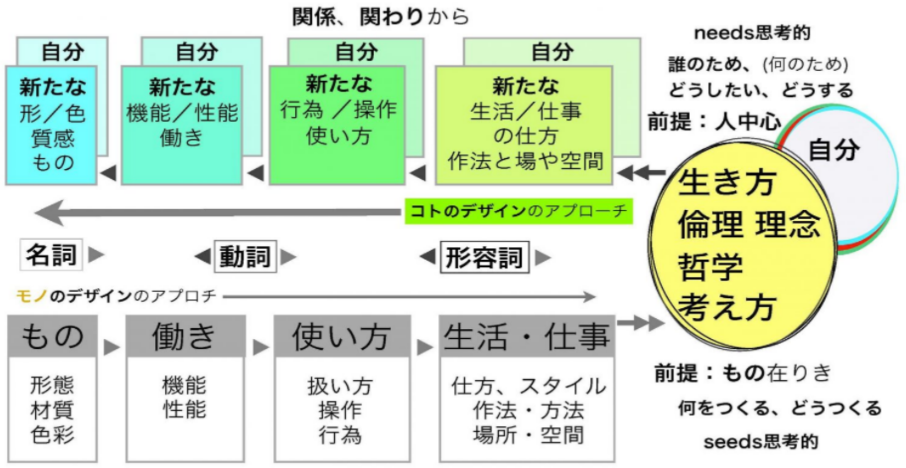


図１ モノのデザインとコトのデザインの違い

モノのデザイン

モノのデザインは、物理的なモノありき、すなわち物や形をつくることが前提になっています。その思考の流れは、物の形や色や素材から入り、機能や性能、そして扱い方から操作性や行為を考える方向にあります。この延長で生活や仕事の仕方、場や空間に思いを巡らしつつ、人の生き方や考え方、倫理や哲学を捉えようとするデザインです。物として何をつくるか、どうつくるかが主眼であり、しいて言うならば、seeds型のデザインということができます。（図１）

コトのデザイン

一方、コトのデザインは、モノのデザインとは逆のアプローチをとり（図１）、人の生き方や理念、考え方や哲学から「誰のために、何のためにデザインするのか」から始めます。そのデザインは、新しい生活や仕事の仕方、新たな場や空間のあり方の提案になっているのかを熟考し、新たな行為や使い方から新しい働きを生み出すための試行であり、必要であれば、プロセスと考え方に応じた物を通りだす考え方です。コトのデザインの思考によって物をつくりだすこともありますが、ものづくりが最大の目的ではありません。空間や行為や働きをつくりだすデザインもあり得ます。すなわち、図案を描いたり、物をつくりだすことには固執しない、造形から解き放たれたデザインです。いわば、needs型のデザインに近いといえるでしょう。

Source.
https://safeology.org/wp/words/needs/

FOSTERING LEADERS OF INNOVATION IN A KNOWLEDGE INTENSIVE SOCIETY

(2) Specific efforts

(i) Promote challenges for young people to develop their individuality

It is necessary to work on strengthening and networking efforts related to entrepreneurship development, such as the formation of a startup ecosystem centered on universities that are central to entrepreneurship education, and to accelerate the fostering of entrepreneurship and the building of an ecosystem infrastructure for Japan as a whole.

In addition, high schools that provide advanced science and mathematics education are required to develop and implement curricula that are not based on the Courses of Study, and to develop experiential and problem-solving learning through experiments, etc., in order to strengthen human resource development that fosters a spirit of inquiry, creativity, and broad intellectual interests.

[Source] Council for Science, Technology and Science, Special Committee on General Policy (2020), "Science and Technology Innovation for Knowledge Intensive Value Creation.
Policy Development - Toward a World Leading Nation through the Realization of Society 5.0 (Final Summary)".

Initiatives of the Division of Advanced Studies in the Interdisciplinary Academic Field

image of human resources to be trained

Leading the creation of innovation

Human Resources for Social Change

Innovation and Entrepreneurship

Focus on

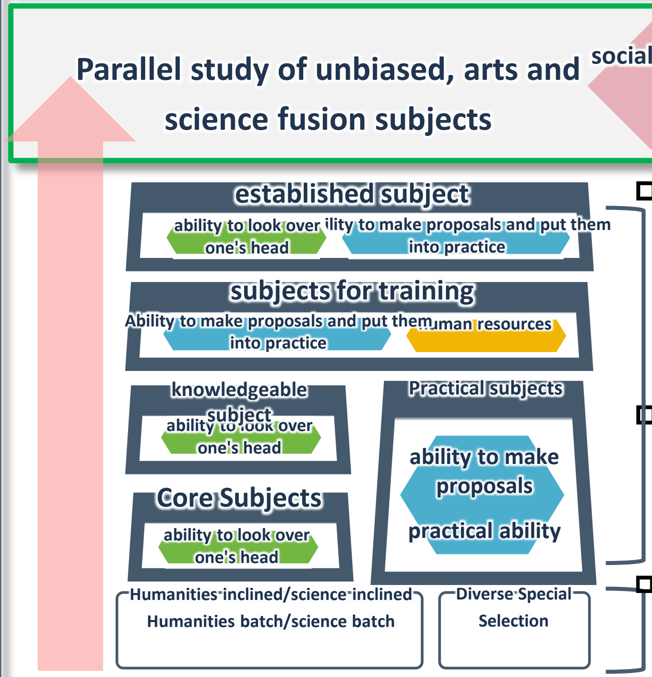
The human resources to be developed in this project

with fused expertise and astute leapfrogging knowledge.

Human Resources for Leading Social Change

Feature 1

Acquire specialized knowledge through a curriculum that integrates the humanities and sciences

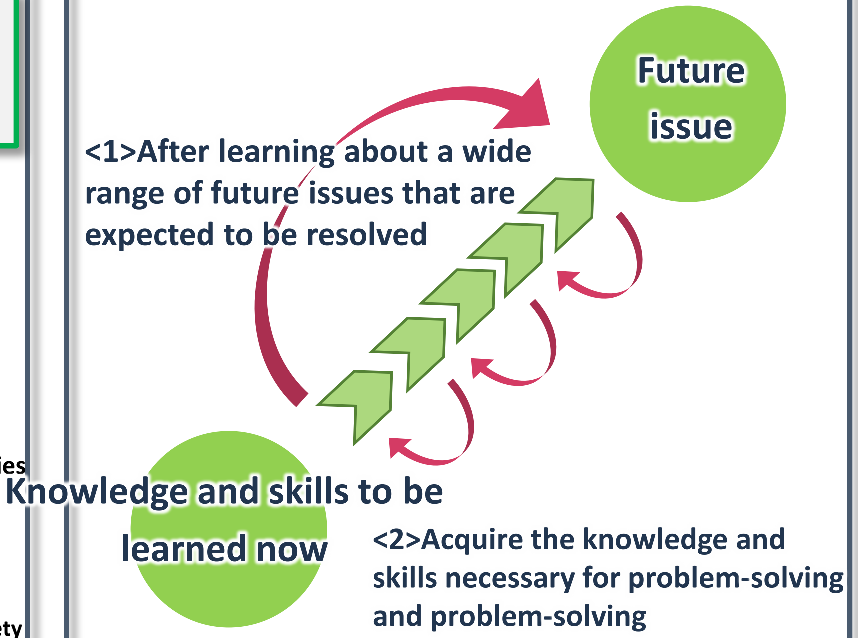


- Fostering innovation orientation and entrepreneurship while traveling back and forth between a wide range of fields
- Students independently design their own learning and cultivate three qualities
- Accepting students of all literary and scientific backgrounds through a variety of entrance examinations

Feature 2

backcasting

Problem-Solving Learning



Unique Classroom Practices in Leading Academies

Entrepreneur Fundamentals" class (1st year, Q1)

[Subject of the class]
Fieldwork in the form of a local training camp with the full cooperation of the Nanao Chamber of Commerce and Industry
Problem-discovery and problem-solving classes that incorporate
As a final summary, a proposal presentation was made to the Nanao Chamber of Commerce and Industry to change its entrepreneurial mind-set as a starting point for future studies in the Leading Schools.
This is a required course that is positioned as the first step.

Business Proposal for Nanao City Officials


Discussion by all participants


Lecture by President Yamazaki

*In 2021, due to the spread of new coronavirus infections, the event was conducted online via Zoom over two days on Saturdays and Sundays.

FOSTERING LEADERS OF INNOVATION IN A KNOWLEDGE INTENSIVE SOCIETY

(2) Specific efforts

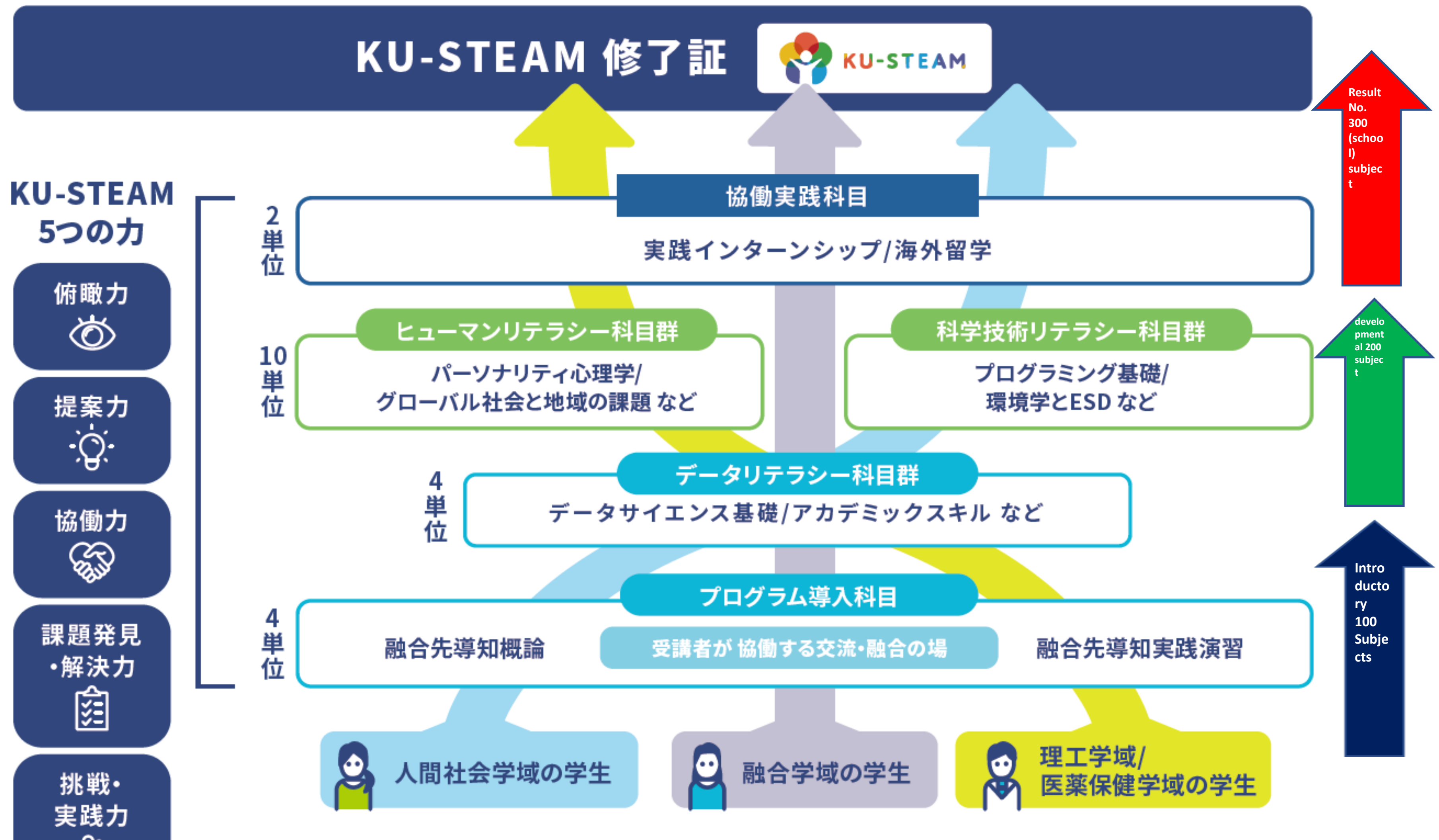
(ii) Promote education that transcends the divisions of arts and sciences to respond quickly to changes in society.

In order to promote the development of human resources that transcends the traditional divisions between the humanities and sciences, society as a whole, including industry, should promote the acquisition of knowledge in probability and statistics, programming, and the basic fields of natural and social sciences, as well as STEAM education, which are the foundation of data science required in various academic fields, while also improving conditions for students who wish to study more advanced content. In addition, it is necessary to create conditions for students who wish to learn more advanced content.

It is also necessary to promote multiple acquisition of different systems of study, such as medicine and philosophy, engineering and economics, etc., and the expansion of mathematical, data science, and AI education to all undergraduate students. Furthermore, through promotion of the introduction of "degree programs that transcend the organizational boundaries of faculties and graduate schools," institutionalized with the aim of enabling universities to exercise mobility at their own discretion and actively engage in cross-faculty education by utilizing internal resources, cross-disciplinary education, such as combining AI with other fields of study, should be promoted. The introduction of "cross-faculty degree programs" is required to promote cross-disciplinary education such as combining AI with other fields.

[Source] Council for Science, Technology and Science, Special Committee on General Policy (2020), "Science and Technology Innovation for Knowledge Intensive Value Creation. Policy Development - Toward a World Leading Nation through the Realization of Society 5.0 (Final Summary)".

Initiatives of the Leading STEAM Human Resource Development Program (KU-STEAM)



「STEAM人材」とは？

Science（科学）、Technology（技術）、Engineering（工学）、Mathematics（数学）といった理数系を学修する「STEM教育」にArt（アート、デザイン）の創造的要素を加え、各領域での学修を実社会での課題解決に活かせる、俯瞰力と実践力を身につけた人材。

FOSTERING LEADERS OF INNOVATION IN A KNOWLEDGE INTENSIVE SOCIETY

(2) Specific efforts

(iii) Establishment of a career system that enables people to work with diverse experience and expertise.

It is expected that human resources with diverse career backgrounds will serve as mediators of "knowledge" and realize a society in which they can play an active role as core players in value creation while making the most of their multiple specialties.

Therefore, it is necessary to develop employment systems and environments that enable the selection of various career paths, such as dual employment, job change, and recurrent education, so that they can be active not only in universities and national research and development corporations, but also in various other institutions, including companies.

[Source] Council for Science, Technology and Science, Special Committee on General Policy (2020), "Science and Technology Innovation for Knowledge Intensive Value Creation.
Policy Development - Toward a World-Leading Nation through the Realization of Society 5.0 (Final Summary)".

FIRST LESSON SUBMISSION

First Lesson assignment (to be submitted via WebClass)

After reading some hand out materials and viewing the on-demand videos, please summarize the following:
"(1) Summarize what is required in a knowledge-intensive society."
Then, "(2) Please summarize what you yourself can do as a professional (researcher, engineer, etc.) in a knowledge-intensive society in the future."
Please summarize the entire report including above (1) and (2),
The entire application must be between 300 and 400 words and submitted via WebClass by a date to be informed.

REFERENCES

REFERENCES

Makoto Gonokami (2017), "Social Transformation toward Society 5.0 (Knowledge Intensive Society) and the Role of Universities," Fiscal System Subcommittee, Council on Fiscal System, Document (2017.10.4)

Makoto Gonokami (2019), "Daigaku no Mirai Chizu - 'Knowledge Intensive Society' wo Tsukuru" (The Future Map of University - Creating a 'Knowledge Intensive Society'), Chikuma Shinsho

General Incorporated Association, Education Alliance Network (2021) "Let's have the power to be a compass for the future! (Koreima NO.13)

Council for Science, Technology and Science, Special Committee on General Policy (2020), "Development of Science and Technology Innovation Policy for Knowledge Intensive Value Creation. - Becoming a World Leader through the Realization of Society 5.0 - (Final Summary)"

Director-General for Science, Technology and Innovation, Cabinet Office, Government of Japan (2020) "Basic Plan for Science, Technology and Innovation - Toward the Realization of Society 5.0

Government (2016), "Basic Plan for Science and Technology."

Government (2021) "Basic Plan for Science, Technology and Innovation.

H Bakhshi, JM Downing, MA Osborne, P Schneider (2017) "The future of skills: employment in 2030"